

Written Analysis

The purpose of this experiment was to compare linear regression using batch and mini-batch gradient descent with no regularization, L1 regularization, and L2 regularization. To conduct this experiment, a combined dataset made up of one Red Wine Quality dataset and one White Wine Quality dataset were used. The data from these datasets was divided into 70% training, 15% validation, and 15% testing subsets.

For both L1 and L2 regularization, the regularization strength was selected using the validation set. The following values were tested: 0.0001, 0.001, 0.01, 0.1, 1.0 and 10.0. For L2 regularization, 0.001 produced the lowest validation MSE of 0.5598. For L1 regularization, 0.001 also performed the best yielding a validation MSE of 0.5600. For both cases, larger regularization strengths significantly increased validation error, seen with values at 1.0 and 10.0.

Both L1 and L2 identified citric acid as the attribute with a weight closest to zero. The weight was -0.00263 with L1 and -0.00446 with L2, so citric acid was eliminated before retraining the reduced model. After doing so, the batch model had a training MSE of 0.5394, 0.5601, and 0.5037. The unregularized mini-batch model similarly produced MSEs of 0.5410, 0.5618, and 0.5065.

Overall, L1 and L2 produced very similar results. Both selected the same α and eliminated the same attribute, although L1 found the citric acid weight closer to zero. Removing citric acid caused almost no change in validation or test MSE, indicating that it contributed relatively little to the model's predictive performance.